



WORN AWAY

Weathering by wind and water can result in the significant erosion of rocks and the sculpting of land surfaces, as seen here in the dramatic landscape of Bryce Canyon, Utah, USA.

WEATHERING AND EROSION

Many rocks are formed below Earth's surface, and when they are exposed by uplift from pressure within Earth's crust or by retreating seas or rivers, they react in many ways with the atmosphere, water, and living organisms. The physical and chemical processes resulting from the interaction of rocks and minerals with the atmosphere are known as weathering. The processes by which rock material

is loosened, dissolved, and transported away is known as erosion. The combination of weathering and erosion wears down Earth's rocky surfaces, layer by layer. Exposed rocks on mountain tops and on the exterior of buildings, for example, are subject to chemical weathering by acid rain and to physical weathering by temperature change and the splintering freeze-thaw effect of ice formation. Bare rock surfaces may also be physically eroded by sand particles carried by the wind. The combined effect of weathering and erosion dissolves some rocks, and reduces others to fragments. As rock debris is broken down and transported by wind, water, and ice, the sediment created becomes increasingly available to life forms, providing important mineral nutrients and a new surface for anchorage and growth.

< LANDSLIDE, RIO DE JANEIRO

Even with a well-developed plant cover, the impact of high rainfall on an area of steep slopes has the potential to cause landscape-altering and even life-threatening occurrences such as avalanches and landslides.



SCIENCE SOIL FORMATION

The production of a soil requires the initial weathering and erosion of the parent rock, which is broken down into small, mineral-bearing particles known as regolith. The addition of humus – organic matter formed from the remains of plants and animals – forms the basis for a soil. The soil in turn becomes the bed in which more life flourishes.

